

AgriTalk: Revolutionizing Farming with a Multilingual Chatbot

Abhishek M V, A Jaya Sreekar, Mohith D M, Susmitha Vekkot, Bhavana V.

*Department of Electronics and Communication Engineering, Amrita School of Engineering,
Bengaluru, Amrita Vishwa Vidyapeetham, India.*

abhishekmv2004@gmail.com, ajsreekar1436@gmail.com, smohith17@gmail.com,
v_susmitha@blr.amrita.edu, v_bhavana@blr.amrita.edu

Abstract—Agriculture plays a vital role in the economy of many nations, especially in regions where a significant portion of the population depends on it for their livelihood. Despite its importance, linguistic and technological barriers often hinder access to accurate agricultural information. This article introduces a multilingual chatbot designed to address agricultural queries efficiently. Leveraging Natural Language Processing and deep learning models, this multi-language chatbot accommodates English, Hindi, Malayalam, Kannada, and Telugu. Since the query is matched with relevant responses using cosine similarity on the sentence-transformers architecture, the system ensures that relevant responses are given. With IndicTrans2 (AI4Bharat), a robust neural machine translation model tailored for Indian languages, the facility provides a single-machine foundation on which translations about 22 Indian languages with subsequent additional features-text-to-speech functionality enhances accessibility to varied users. The system design, data preprocessing, implementation, and evaluation metrics are covered in this study, which also shows how it might help farmers and other agricultural stakeholders.

Keywords—NLP, multilingual systems, sentence transformers, cosine similarity, agricultural chatbots, and accessibility.

I. INTRODUCTION

The global economy depends heavily on agriculture, especially in areas where a sizable section of the population depends on it for living. Despite its critical importance, the industry nevertheless faces challenges such as restricted access to accurate and timely information. Agricultural research often covers a wide range of topics, such as weather forecasting, market trends, crop management, and pest control. It is essential to be able to answer these questions accurately and contextually to empower farmers and increase output. However, the main challenges remain the diversity of regional languages and linguistic hurdles in many rural areas. To address these challenges, multilingual tools are both socially and technologically required [1-3].

Natural language processing (NLP) and deep learning have emerged as revolutionary technologies for developing intelligent systems, such as chatbots, for a variety of application domains. Multilingual chatbots are very useful for agricultural advising services because they can process requests in multiple languages [4-5]. However, existing algorithms sometimes struggle to answer complex queries,

cannot ensure the accuracy of generated responses, and are unable to adequately account for language diversity [6-8]. These disadvantages emphasize the need for robust multilingual chatbots that can bridge the linguistic and informational limitations that farmers face.

The application of NLP to the development of domain-specific chatbots has been the subject of numerous studies. Using English datasets and simple keyword-matching algorithms, one approach created a chatbot for agricultural consulting services; however, it struggled with semantics and did not support regional languages. To address multilingual queries, another study included translation tools; nonetheless, the translations frequently produced inaccurate results. On the other hand, by guaranteeing precise and contextual translations across 22 Indian languages, this work makes use of IndicTrans2 (AI4Bharat), a cutting-edge multilingual neural machine translation model, to successfully address linguistic variety. Deep learning methods have been used to develop chatbots in other fields, but they have not addressed the linguistic and contextual difficulties that the agricultural industry faces.

The lack of precise multilingual support for agricultural inquiries and the requirement for an effective method to match queries with pertinent answers in a big data set are the two main issues that this study aims to address to close these gaps. The proposed work presents a multilingual chatbot that works for Telugu, Hindi, Malayalam, Kannada, and English. The system uses cosine similarity-based matcher-like queries with relevant responses and state-of-the-art sentence transformers for creating semantic embeddings. High-quality, contextually sensitive translations specific to Indian languages are guaranteed by the IndicTrans2's translation capabilities. The following are the contributions made by this work:

- creation of a multilingual chatbot specifically designed to answer questions about agriculture and consider the language diversity of farmers.
- Integration of modern NLP techniques, including sentence-transformers, to assure semantic understanding and correctness.
- Incorporation of text-to-speech capability to boost accessibility for non-literate users.

- Incorporation of IndicTrans2 for accurate and nuanced translations across supported languages.
- Comprehensive examination of the system's performance across numerous languages and query types.

The rest of this paper is organized as follows: Section II addresses the system design and implementation, including details regarding data preprocessing, embedding generation, and query matching. Section III discusses the evaluation technique and findings, focusing on relevance scores, user comments, and translation accuracy. Section IV discusses the limitations of the existing work and outlines potential future improvements. Finally, Section V closes the study by summarizing the contributions and underlining its broader implications [9], [10].

II. LITERATURE SURVEY

Numerous chatbot solutions have been investigated by researchers to address difficulties in agriculture. With strong usage ratings, Arora et al. [1] demonstrated the potential of a generative neural network-based chatbot that can perform tasks connected to crops, weather forecasting, and pest control. Retrieval-augmented generation (RAG) techniques were utilized by Singh et al. [2] to build Farmer Chat, a multilingual chatbot that achieved a 75% accuracy rate in over 300,000 user queries, considerably enhancing user engagement.

Using the Kisan Call Center dataset, Momaya et al. [3] built a RASA-based chatbot that relates to WhatsApp. It reduced the quantity of unanswered inquiries by 40% and attained an intent classification accuracy of 96.1%. Jain et al. [4] developed a multilingual chatbot utilizing NLP to respond to inquiries on fertilizer, pest management, and soil health, emphasizing the technology's capacity to overcome communication barriers in rural areas.

Mishra et al. [5] demonstrated the efficacy of large language models by utilizing ChatGPT to address cross-lingual agricultural issues, achieving over 90% accuracy in multilingual datasets. Alimkar et al. [6] developed Kisan Mitra, a chatbot that provides weather updates, guidance on crop selection, and information on government regulations, emphasizing its utility for impoverished farmers. Niranjana et al. [7] demonstrated the efficacy of artificial neural networks (ANN) in managing historical data by developing a chatbot that responds to inquiries about crop management and pest control. Bhatia et al. [8] introduced a similarity search-based chatbot that utilizes natural language processing techniques to enhance question resolution in domains such as soil health and fertilizer recommendations.

Prakash et al. [9] utilized NLP and machine vision to create a chatbot that can answer to text and image-based agricultural queries and identify plant illnesses with 85% accuracy. A scalable chatbot framework for crop planning and fertilizer

prediction was proposed by Maduri et al. [10] and was lauded for its intuitive design. Although it lacked complicated NLP capabilities and performance measurements, Anjali et al. [11] focused on a voice-activated chatbot designed for rural farmers that made communication simple. An NLP-driven chatbot was constructed by Biswas et al. [12] for real-time soil management query answer; however, its usability was impeded by its limited language support.

A chatbot optimized for mobile devices was built by Nikhil Sharma et al. [13], who prioritized accessibility above sophisticated natural language processing. Despite its lack of sophistication, the system's objective was to utilize rudimentary machine learning techniques to solve farming challenges. On the other hand, a structured Q&A framework for agricultural queries was presented in [14]. Although effective for certain queries, the system was less adaptable than more dynamic methods due to its lack of conversational flexibility and sophisticated metrics.

Using the CTC approach with XLSR Wav2Vec2.0, Akhilesh et al. [15] demonstrated cross-lingual transfer learning with a WER of 0.58 for Tamil speech recognition, but they had trouble scaling to other dialects. Despite scarce resources, Vancha et al. [16] helped preserve the Sourashtra language by creating a Kaldi-based speech recognition system that achieved a WER of 5.5% on 2000 utterances. Using an Arduino Mega 2560 and GSM 800L, Shrivastava et al. [17] suggested an automated watering system that relies on sensor data to send out email and SMS alerts, but it lacks sophisticated predictive analytics.

III. SYSTEM ARCHITECTURE

The system architecture of the proposed multilingual chatbot, as shown in Fig.1, is designed to handle varied agricultural queries efficiently and accurately. The architecture is made of multiple interconnected modules that enable the chatbot to analyze user queries, execute translations, build embeddings, match inquiries to relevant answers, and offer both written and voice responses. Such an approach coincides with techniques mentioned in [11], where multilingual processing and modular chatbot designs were stressed to accommodate linguistic and functional variety.

A. System Flow

The following steps outline the flow of the system:

a) *User-Submitted Queries*: When a user submits a question via the chatbot interface, the system starts up. Any of the accessible languages-English, Hindi, Malayalam, Kannada, or Telugu may be used for this input. The inquiry is recorded by the interface as text and transmitted to other modules for processing. Effective query processing in multilingual input systems depends on seamless user

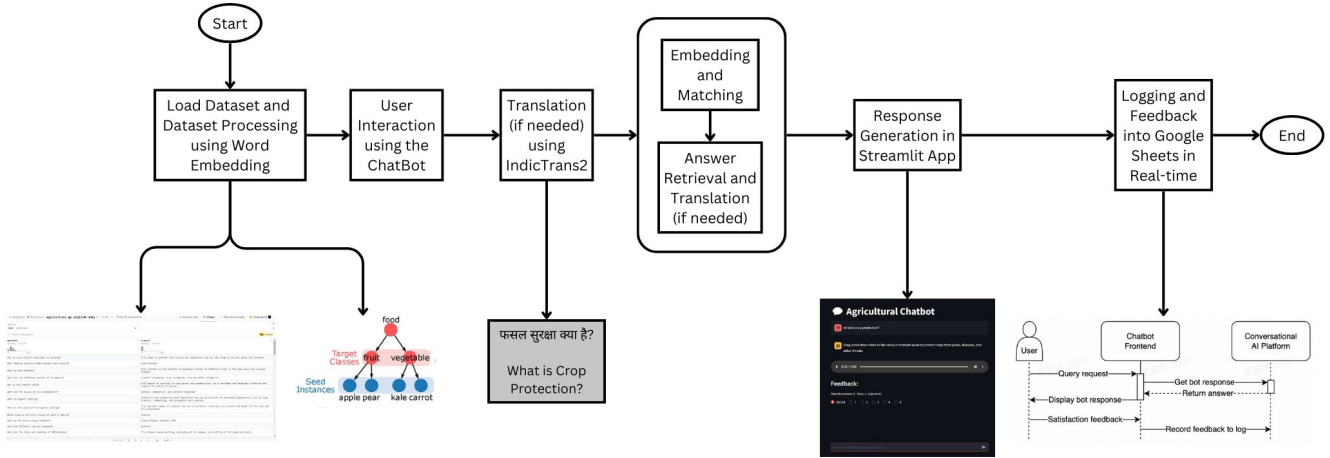


Fig. 1. System Architecture of the Proposed Model

interaction because it directly affects response efficiency and accuracy, guaranteeing a consistent experience across multiple languages.

b) Identification and Translation of Languages: The first module applies language detection technology to identify the query's language. If the query is not in English, *IndicTrans2* (AI4Bharat) is used to turn it into English. Since the dataset and embedding models are trained in English, this translation stage provides language consistency for all operation. By focusing on the nuances of Indian languages, *IndicTrans2* boosts translation quality and produces exact and contextually sensitive translations.

c) Embedding Generation of Queries: Once translated, the query is given to the embedding producing module. This module employs the sentence-transformers model *paraphrase-multilingual-MiniLM-L12-v2*. to convert the query into a dense numerical vector. This vector reflects the query's semantic meaning, enabling accurate matching with similar searches in the dataset.

d) Calculating Similarities: Precomputed embeddings of every question in the dataset are compared to the generated query embedding. Each dataset question's relevance to the user's inquiry is examined using a metric called cosine similarity. The most pertinent question is found by sorting the similarity scores.

e) Decision: Found a Relevant Answer? The system decides if the top match's similarity score is higher than a predetermined threshold (e.g., 0.7). The matched response from the dataset is picked, if the threshold is attained. If not, the system logs the query for more study and confesses that it was unable to locate a pertinent response.

f) Generate Text and Audio Response: The system produces a text response if it identifies a pertinent response. Additionally, Google Text-to-Speech (gTTS) is employed to deliver an audio version of the response for greater accessibility. This stage guarantees that clients with varying levels of literacy may utilize the system.

g) Apologize and Log Feedback: The system tells the user that it is unable to deliver an appropriate response if no relevant answer is detected. The inquiry is logged for further study, including its context and the system's judgment. The system's continual progress depends on this feedback mechanism.

h) Response Delivered: Finally, the user receives the text and audio response via the chatbot interface. By doing this, the question resolution process is performed, guaranteeing that the user will acquire the greatest response or recognition.

B. Key Features

Several major aspects are added into the system architecture to boost its usability and performance:

- **Multilingual Support:** The system's multilingual support provides accessibility and inclusion for a wide range of users. For Indian languages, *IndicTrans2* (AI4Bharat) powers translation capabilities, enabling precise and culturally meaningful translations
- **Semantic Understanding:** The system makes use of powerful sentence-transformer models to assure precise query interpretation and matching.
- **Audio Accessibility:** For users with low literacy levels, the text-to-speech tool delivers an audio answer.
- **Continuous Improvement:** To uncover system problems and direct future changes, user feedback is logged.

The chatbot is a dependable device for answering agricultural queries in a range of languages and geographical areas thanks to its sturdy and modular architecture, which assures scalability and flexibility.

IV. EVALUATION METRICS

Table I presents the chatbot's performance numbers, which give a basic impression of its effectiveness, precision [18-21], and user contentment. The system's proficiency in man-aging multilingual queries is evidenced by critical metrics like translation accuracy, relevance score, response times, and language detection accuracy. Metrics like error rate and feedback rating also reflect user perceptions and opportunities for improvement. In the sections that follow, each of these indications will be addressed in further detail to give a full sense of the system's performance and prospective changes.

a) *Language Detection Accuracy and Report:* To effectively direct multilingual questions, the chatbot must be able to distinguish user query languages with perfect accuracy (1.0) and precision, recall, and an F1-score of 1.0.

b) *Average Relevance Score:* The chatbot successfully matches responses with user requests to boost customer happiness, as demonstrated by its relevance score of 0.91.

c) *Error Rate:* A 19.35% mistake rate reveals the percentage of erroneous responses and points out areas in which certain subjects need attention.

d) *Response Time Metrics:*

- **Average:** 2.37 seconds, ensuring efficient response generation.
- **Minimum:** 1.03 seconds, demonstrating quick response potential.
- **Maximum:** 7.13 seconds, exhibiting sporadic lags.

e) *Fallback Rate:* The frequency of generic responses coming from unmatched queries in the dataset is reflected by a fallback rate of 11.29%.

f) *Translation Accuracy:* With a translation accuracy of 1.0, the chatbot enables continuous multilingual discussions and user confidence.

g) *Average Feedback Rating:* Above-average customer contentment and views of efficacy are represented by a rating of 3.53/5.

A. Utility of These Metrics for the Chatbot

- **Improved Multilingual Support:** Validates the chatbot's proficiency in handling different languages.
- **Feedback for Training:** Finds regions that need modifications to the dataset or model.
- **User-Centric Performance:** Highlights user happiness to guide iterative improvements.
- **Operational Efficiency:** Ensures reaction times stay engaging under varying loads.

V. RESULTS AND DISCUSSION

The technology indicates tremendous potential in addressing the information demands of farmers in linguistically varied locations. By integrating modern NLP techniques and deep learning models, the chatbot gives correct and context-aware responses. However, hurdles remain in increasing translation accuracy and expanding the dataset to include more domain-specific questions.

The chatbot successfully gives a text response coupled with an audio version of the answer as seen Fig. 2, proving its capacity to fetch relevant information from the dataset. Additionally, the interface incorporates a feedback mechanism, allowing users to score the correctness of the response, which may be utilized for refining the system.

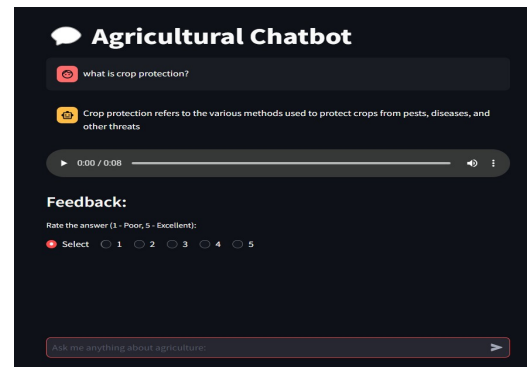


Fig. 2. Chatbot providing text and audio responses with feedback mechanism.

The chatbot effectively handles multiple questions sequentially and displays accurate responses in text and audio formats as shown Fig. 3. User feedback for the responses is recorded in real time and logged into a Google Sheet, ensuring that all interactions are tracked for analysis. The system successfully handles and responds to queries in multiple languages, such as Hindi, Malayalam, Telugu, and Kannada as shown Fig. 4. The chatbot adapts to the language of the input and provides appropriate text and audio responses, showcasing its versatility in handling multiple languages.

The user interface and real-time statistics of the multilingual chatbot for agricultural support are shown in Fig. 5. The interface, which is displayed on the left, allows the users to ask questions in various languages and get text and voice responses in return. It also lets users rate the chatbot's responses and verify that the translations are accurate. The conversation logs, which are shown on the right side of the figure, methodically record information like the user's query, the chatbot's response, the languages that were detected and those that were spoken, relevance scores, response times (average, minimum, and maximum), fallback usage,

TABLE I
CHATBOT PERFORMANCE METRICS

Metric	Value	Inference
Language Detection Accuracy	1.0	The chatbot successfully detected the language for all queries correctly.
Average Relevance Score	0.91	Indicates high alignment between user queries and chatbot responses.
Error Rate	19.35%	Reflects that approximately 19% of the chatbot's responses were incorrect.
Response Time Metrics	Avg: 2.37s, Min: 1.03s, Max: 7.13s	Highlights the efficiency of the chatbot in delivering responses.
Fallback Rate	11.29%	Measures the percentage of queries that resulted in fallback responses.
Translation Accuracy	1.0	Demonstrates perfect accuracy in translating queries to/from supported languages.
Average Feedback Rating	3.53	Suggests users found the chatbot's performance to be above average but with room for improvement.
Language Detection Report	Precision, Recall, F1-Score: 1.0	Confirms the chatbot handles all supported languages with perfect precision and recall.
Overall Accuracy	80.65%	Represents the percentage of correct responses provided by the chatbot out of total queries.

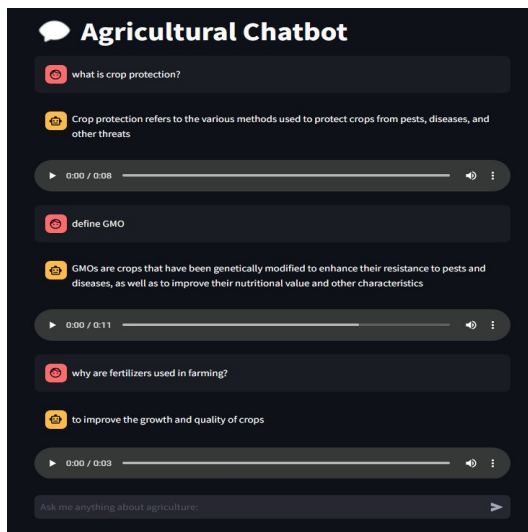


Fig. 3. Chatbot handling multiple questions and updating feedback in real-time.

translation accuracy, feedback ratings, and timestamps. Critical performance indicators such as Language Detection Accuracy, Average Relevance Score, Error Rate, Fallback Rate, and Average Feedback Rating are provided by these logs, which are taken from real-time chatbot conversation recordings that are kept in a Google Sheet. It is crucial to keep an eye on these indicators to evaluate system performance in various languages, guarantee accurate language detection, timely responses, and user satisfaction.

The team can overcome linguistic barriers and provide timely, context-specific solutions by identifying areas for improvement, optimizing translation accuracy, improving response relevance, and improving the chatbot's ability to efficiently address the agricultural needs of multilingual users.

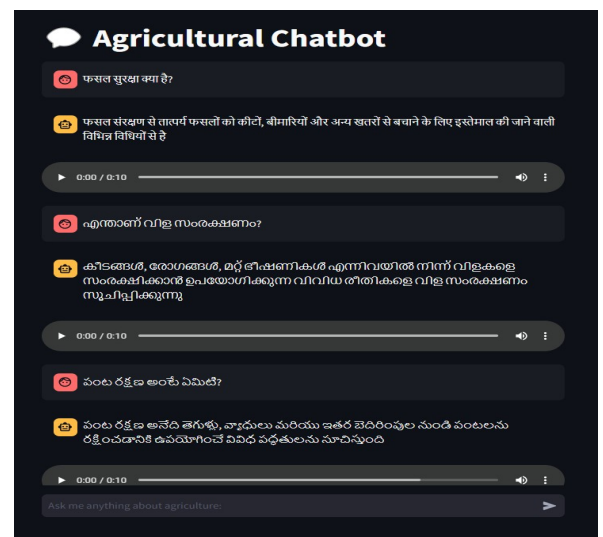


Fig. 4. Chatbot handling multilingual queries and providing corresponding responses.

The multilingual chatbot system that was built has demonstrated great promise in meeting farmers' needs for agricultural information in linguistically diverse areas. The system efficiently analyzes user queries and provides context-aware responses in several languages by leveraging deep learning models and sophisticated natural language processing algorithms. Its applicability in real-world situations is highlighted by its broad range of themes, including crop management, pest control, soil health, and government schemes. Furthermore, accessibility is improved by including both textual and audio responses, especially for farmers with low literacy or technical proficiency. Although the system effectively fills in the linguistic and informational gaps, there are still issues with increasing the accuracy of translations for regionally unique terms and growing the dataset to accommodate regional farming methods. The system's ability to serve as a reliable and comprehensive tool for farmers will be enhanced by these advancements.

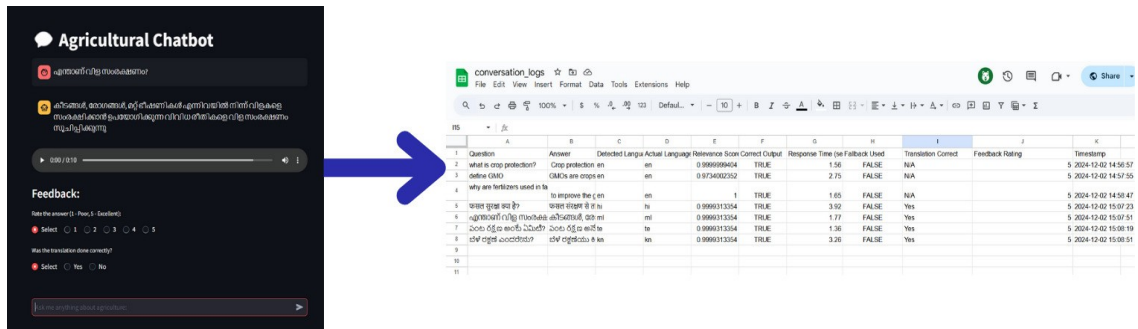


Fig. 5. Conversation logs with User Feedback stored in real-time into Google Sheets.

VI. CONCLUSION AND FUTURE WORK

This study presents a powerful multilingual chatbot that can respond to inquiries about agriculture in several regional languages. For a wide range of users, including farmers in rural and linguistically varied areas, the chatbot guarantees effective and precise query resolution by combining cutting-edge NLP models, translation tools, and embedding techniques. The system is engagingly user-friendly, especially for non-literate users, because it responds both textually and in audio. This innovative model creatively bridges the gap between language barriers to democratize access to vital agricultural knowledge such as crop husbandry, pest management, government programs, and weather reports. The bot's scalable and flexible design thus makes it an exceptionally valuable tool towards increasing agricultural productivity and better farmer decision-making.

The next stage of the study is to extend the database with questions and answers from core localities that offer essential agricultural perspectives and practices. Such improved adaptation will allow the chatbot to provide specific support corresponding to the unique agriculture contexts and needs of geographical areas. Analyses will be conducted in various languages, probabilities of translation models for inclusion of linguistic accuracy and integral contextual integrity in the response across other different languages. Other priorities include exploring the use of visual and sensor data, such as identifying diseases through crop image analysis to improve responsiveness and efficiency to support farmers worldwide.

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